

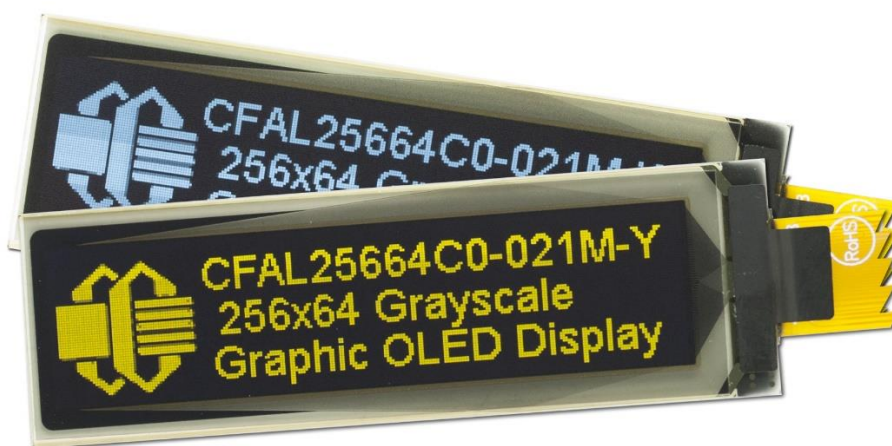


Crystallfontz America, Inc.
<https://www.crystallfontz.com>

CFAL25664C0-021M-[W|Y] OLED Display Module
Datasheet Release Date: 2020-03-16



OLED DISPLAY MODULE DATASHEET



Datasheet Release Date 2020-03-16
for
CFAL25664C0-021M-[W|Y]

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1. General Information

Datasheet Revision History

Datasheet Release: 2020-03-16

Datasheet for the CFAL25664C0-021M-[W|Y] OLED graphic display module.

Product Change Notifications

You can check for or subscribe to [Part Change Notices](#) for this display module on our website.

Variations

Slight variations between lots are normal (e.g., contrast, color, or intensity).

Volatility

This display module has volatile memory.

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2. Module Description

This is a white [W] or yellow [Y] graphic OLED passive matrix display module with a 4-bit grayscale dot matrix. The display is ultra-thin and lightweight. This display has a built-in Solomon Systech SSD1362 controller.

Please see the [SSD1362 datasheet](#) for further reference.

3. Features

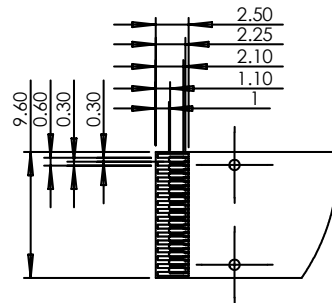
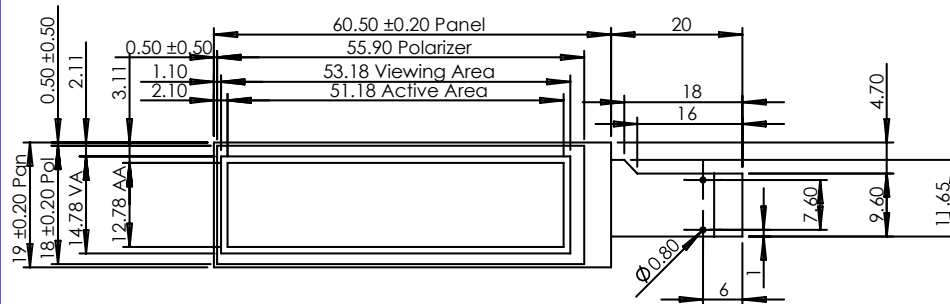
- 256*64 Dot Matrix
- Built-in Controller: SSD1362 or equivalent
- +3v Logic Supply Voltage
- +12v for Display Voltage
- Anti-glare Polarized
- 1/64 Duty
- Interface: 6800, 8080, 3- and 4-wireSPI, I2C
- 2.1" diagonal

4. Mechanical Data

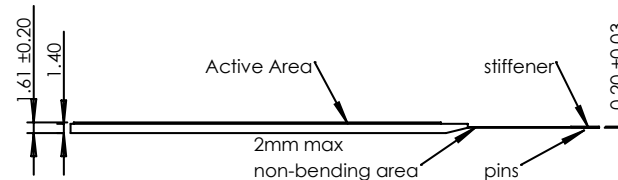
Item	Specification (mm)	Specification (in, reference)
Overall Dimensions (incl. tail)	80.50 x 19.00 x 1.61	3.17 x 0.748 x 0.068
Display Dimensions	60.50 x 19.00 x 1.61	2.38 x 0.748 x 0.068
Active Area	51.18 x 12.78	2.01 x 0.503
Pixel Size	0.18 x 0.18	0.007 x 0.007
Pixel Pitch	0.20 x 0.20	0.008 x 0.008
Weight (Typical)	3.5 grams	0.12 ounces



5. Mechanical Drawing



FPC Section View



Pin	Function
1	NC
2	VOLED
3	VP
4	VCOMH
5	VLogic
6	VOLED
7	IREF
8	VOLED
9	NC
10	VSS
11	VSS
12	VCI
13	BS0
14	BS1
15	VCI
16	BS2
17	CS
18	RES
19	D/C
20	R/W
21	E
22	D0
23	D1
24	D2
25	D3
26	D4
27	D5
28	D6
29	D7
30	NC
31	VOLED

Units: millimeters
Tolerance: ± 0.3



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Part Number:

CFAL25664C0-021M-[W|Y]

Date:

2020-03-10

Filename:

CFAL25664C0-021M-W mm.pdf

Revision:

v1.0

Web:

www.crystalfontz.com/product/cfal25664c0021mw

Sheet:

1 of 1



6. Interface Pin Function

PIN No.	Symbol	Function
1	NC	The NC pins between function pins are reserved for compatible and flexible design.
2	V _{OLED}	Power Supply for Display Connect to an external source to power display. Also called V _{CC} .
3	V _P	Segment Pre-Charge Voltage Reference Connect a 1.0 μF capacitor between this pin and ground. No external power is connected to this pin.
4	V _{COMH}	COM Signal Deselect Voltage Connect a 4.7 μF capacitor between this pin and ground. No external power is connected to this pin.
5	V _{Logic}	Power Supply for Logic Connect a 1.0 μF capacitor between this pin and ground. V _{DD} can be supplied externally (1.65v-2.6v) or regulated internally by V _{CI} when V _{CI} is greater than 2.6v. Also called V _{DD} .
6	V _{OLED}	Power Supply for Display Connect to an external source to power display. Also called V _{CC} .
7	I _{REF}	Segment Output Current Reference For an external I _{REF} , connect a resistor between this pin and V _{SS} to maintain a current around 18.75 μA (~510 kΩ)
8	V _{OLED}	Power Supply for Display Connect to an external source to power display. Also called V _{CC} .
9	NC	The NC pins between function pins are reserved for compatible and flexible design.
10	V _{SS}	Ground. Connect to an external ground.
11	V _{SS}	Ground. Connect to an external ground.
12	V _{CI}	Low Voltage Power Supply for Operation V _{CI} must be greater than or equal to V _{DD} and V _{DDIO}
13, 14, and 16	BS0, BS1, and BS2	BS[2:0] Interface
		000 4-wire SPI
		001 3-wire SPI
		110 8-bit 8080 parallel
		100 8-bit 6800 parallel
		010 I2C
		1 is connected to V _{CI} . 0 is connected to V _{SS} .
15	V _{CI}	Low Voltage Power Supply V _{CI} must be greater than or equal to V _{DD} and V _{DDIO}
17	CS	Chip Select This chip is enabled for MCU communication when CS is pulled low. For I2C, connect to V _{SS}
18	RES	Power Reset for Controller and Driver When the pin is low, initialization of the chip is executed. Keep high during normal operation.
19	D/C SAO	Data/Command When high, data bus is treated as display data. When low, data bus is treated as command data. I2C SAO Distinguish different addresses. SPI 3-Wire Connect to V _{SS} .
20	R/W WR	6800 Parallel Read/Write Read/Write selection input. Read when this pin is pulled high, and write mode when this pin is pulled low and the chip is selected and enabled. 8080 Parallel Write Write input, data write when this pin is pulled low and the chip is selected. Tie this pin LOW if not used



PIN No.	Symbol	Function
21	E RD	6800 Parallel Enable Read/Write operation is enabled when this pin is pulled high and the chip is selected. 8080 Parallel Read Read operation is initiated when this pin is pulled low and the chip is selected.
22-29	D0-D7 SCL, SDA	Parallel Host Data Input/Output Bus 8-bit bidirectional data bus. I2C SCL and SDA D0 (pin 22) is the serial clock input. Tie D1 and D2 together to serve as SDA. SPI MOSI and SCLK D0 serves as SCL. D1 serves as MOSI. Tie unused pins low.
30	NC	The NC pins between function pins are reserved for compatible and flexible design.
31	V _{OLED}	Power Supply for Display Connect to an external source to power display. Also called V _{CC} .

7. Schematic

Recommended components:

C1 and C2: 4.7 μ F

C3, C4, and C5: 1 μ F

Voltage at I_{REF} = V_{OLED} - 2.4v

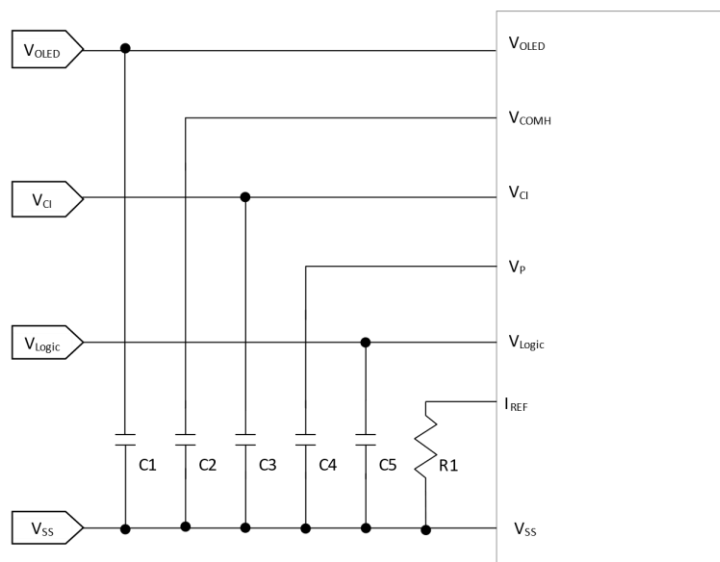
For V_{OLED} = 12v, I_{REF} = 18.75 μ A

$$R1 = \frac{(V_{IREF} - V_{SS})}{I_{REF}}$$

$$R1 = \frac{(12 - 2.4 - 0)}{18.75}$$

$$R1 = \frac{(V_{IREF} - V_{SS})}{I_{REF}}$$

$$R1 = 510k\Omega$$





8. Absolute Maximum Ratings

Parameter	Symbol	Min	Max	Unit
Supply Voltage for Logic	V _{Logic}	-0.5	2.75	V
Supply Voltage for Display	V _{OLED}	-0.5	21	V
Supply Voltage for Operation	V _{CI}	-0.5	5.5	V
Operating Temperature	T _{OP}	-40	80	°C
Storage Temperature	T _{STG}	-40	85	°C

Notes:

- (1) Ratings based on condition that V_{SS}=0v
- (2) These are stress ratings only. Extended exposure to the absolute maximum ratings listed above may affect device reliability or cause permanent damage. Functional operation should be restricted to the limits in the Electrical Characteristics table below

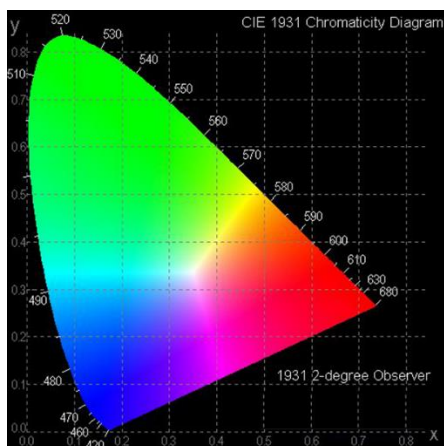
9. Electrical Characteristics

Item	Symbol	Condition	Min	Typ	Max	Unit
Supply Voltage for Operation	V _{CI}	-	2.8	3.0	3.3	V
Supply Voltage for Display	V _{OLED}	-	11.5	12.0	12.5	V
High-level Input	V _{IH}	-	0.8 x V _{CI}	-	V _{CI}	V
Low-level Input	V _{IL}	-	V _{SS}	-	0.2 x V _{CI}	V
High-level Output	V _{OH}	-	0.9 x V _{CI}	-	V _{CI}	V
Low-level Output	V _{OL}	-	V _{SS}	-	0.1 x V _{CI}	V
50% Checkerboard operating current	I _{CC}	V _{OLED} =12V		15	30	mA
Operating lifetime		T _a =25°C		20,000		Hours

OLED lifetime is defined as the amount of time before the luminance has decayed to less than 50% of its initial value. This number is estimated using accelerated conditions to estimate for products under normal use conditions. Screen saving mode will extend OLED lifetime.

10. Optical Characteristics

Item	Symbol	Condition	Min	Typ	Max	Unit
View Angle - Vertical	θ	-	160	-	-	deg
View Angle - Horizontal	φ	-	160	-	-	deg
Contrast Ratio	CR	Dark	-	2,000:1	-	-
Response Time	T _{Rise}	-	-	10	-	μs
	T _{Fall}	-	-	10	-	μs
Brightness (White)		Use typical voltages	80	100	-	cd/m2
CIE _x (White)	X	(CIE1931)	0.26	0.28	0.30	-
CIE _y (White)	Y	(CIE1931)	0.30	0.32	0.34	-
Brightness (Yellow)		Use typical voltages	100	120	-	cd/m2
CIE _x (Yellow)	X	(CIE1931)	0.45	0.47	0.49	-
CIE _y (Yellow)	Y	(CIE1931)	0.48	0.50	0.52	-



11. OLED Module Precautions

The precautions below should be followed when using OLED modules to help ensure personal safety, module performance, and compliance of environmental regulations.

11.1. Modules

- Avoid applying excessive shocks to module or making any alterations or modifications to it.
- Do not make extra holes on the printed circuit board, modify its shape or change the components of OLED display module.
- Do not disassemble the OLED display module.
- Do not operate the OLED display module above the absolute maximum rating.
- Do not drop, bend or twist the OLED display module.
- Soldering: only to the I/O terminals.
- Store in an anti-static electricity container and clean environment.
- It is common to use the "screen saver" to extend the lifetime of the OLED display module.
 - Do not use the fixed information for long periods of time in real application.
 - Do not use fixed information in OLED panel for long periods of time to extend "screen burn" effect time.
- Crystallfontz has the right to change the passive components, including R2 and R3 adjust resistors. (Resistors, capacitors and other passive components will have different appearance and color caused by the different supplier.)
- Crystallfontz have the right to change the PCB Rev. (In order to satisfy the supplying stability, management optimization and the best product performance, etc., under the premise of not affecting the electrical characteristics and external dimensions, Crystallfontz has the right to modify the version.).

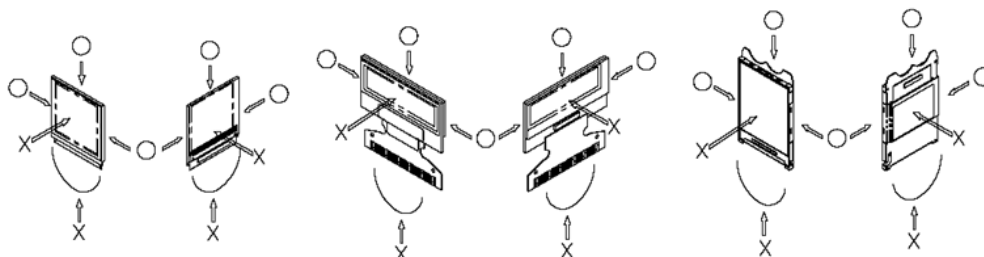
11.2. Handling Precautions

- Since the display panel is made of glass, do not apply mechanical impacts such as dropping from a high position.
- If the display panel is accidentally broken, and the internal organic substance leaks out, be careful not to inhale or touch the organic substance.
- If pressure is applied to the display surface or its neighborhood of the OLED display module, the cell structure may be damaged, so be careful not to apply pressure to these sections.
- The polarizer covering the surface of the OLED display module is soft and can be easily scratched. Please be careful when handling the OLED display module.
- Clean the surface of the polarizer covering the OLED display module if it becomes soiled using following adhesion tape.
 - Scotch Mending Tape No. 810 or an equivalent
 - Never breathe the soiled surface or wipe the surface using a cloth containing solvent such as ethyl alcohol, since the surface of the polarizer will become cloudy.
 - The following liquids/solvents may spoil the polarizer:
 - Water
 - Ketone



– Aromatic Solvents

- Hold the OLED display module very carefully when placing the OLED display module into the system housing.
- Do not apply excessive stress or pressure to the OLED display module. And, do not over bend the film with electrode pattern layouts. These stresses will influence the display performance. Also, be sure to secure the sufficient rigidity for the outer cases.



- Do not apply stress to the LSI chips and the surrounding molded sections.
- Do not disassemble or modify the OLED display module.
- Do not apply input signals while the logic power is off.
- Pay sufficient attention to the working environments when handling the OLED display module to prevent occurrence of element breakage accidents by static electricity.
 - Be sure to make human body grounding when handling OLED display modules.
 - Be sure to ground tools to use for assembly such as soldering irons.
 - To suppress generation of static electricity, avoid carrying out assembly work under dry environments.
 - Protective film is being applied to the surface of the display panel of the OLED display module. Be careful since static electricity may be generated when exfoliating the protective film.
- Protection film is being applied to the surface of the display panel and removes the protection film before assembling it. At this time, if the OLED display module has been stored for a long period of time, residue adhesive material of the protection film may remain on the surface of the display panel after the film has been removed. In such a case, remove the residue material by the method discussed above.
- If electric current is applied when the OLED display module is being dewed or when it is placed under high humidity environments, the electrodes may become corroded. If this happens proceed with caution when handling the OLED display module.

11.3. Storage Precautions

- When storing the OLED display modules put them in static electricity preventive bags to avoid exposure to direct sunlight and fluorescent lamps. Also avoid high temperature and high humidity environments and low temperatures (less than 0°C) environments. (We recommend you store these modules in the packaged state when they were shipped from Crystallfontz). Be careful not to let water drops adhere to the packages or bags, and do not let dew gather on them.
- If electric current is applied when water drops are adhering to the surface of the OLED display module the OLED display module may have become dewed. If a dewed OLED display module is placed under high humidity environments it may cause the electrodes to become corroded. If this happens proceed with caution when handling the OLED display module.

11.4. Designing Precautions

- The absolute maximum ratings are the ratings that cannot be exceeded for OLED display module. If these values are exceeded, panel damage may happen.
- To prevent occurrence of malfunctioning by noise pay attention to satisfy the V_{IL} and V_{IH} specifications and, at the same time, to make the signal line cable as short as possible.
- We recommend that you install excess current preventive unit (fuses, etc.) to the power circuit (V_{DD}). (Recommend value: 0.5A)
- Pay sufficient attention to avoid occurrence of mutual noise interference with the neighboring devices.
- As for EMI, take necessary measures on the equipment side.
- When fastening the OLED display module, fasten the external plastic housing section.



- If the power supply to the OLED display module is forcibly shut down, by such errors as taking out the main battery while the OLED display panel is in operation, we cannot guarantee the quality of this OLED display module.
 - Connection (contact) to any other potential than the above may lead to rupture of the IC.

11.5. Disposing Precautions

- Request the qualified companies to handle the industrial wastes when disposing of the OLED display modules. Or, when burning them, be sure to observe the environmental and hygienic laws and regulations.

11.6. Other Precautions

- When an OLED display module is operated for a long period of time with a fixed pattern, the fixed pattern may remain as an after image or a slight contrast deviation may occur.
 - If the operation is interrupted and left unused for a while, normal state can be restored.
 - This will not cause a problem in the reliability of the module.
- To protect the OLED display module from performance drops by static electricity rapture, etc., do not touch the following sections whenever possible while handling the OLED display modules:
 - Pins and electrodes
 - Pattern layouts such as the TCP & FPC
- With this OLED display module, the OLED driver is being exposed. Generally speaking, semiconductor elements change their characteristics when light is radiated according to the principle of the solar battery. Consequently, if this OLED driver is exposed to light, malfunctioning may occur.
 - Design the product and installation method so that the OLED driver may be shielded from light in actual usage.
 - Design the product and installation method so that the OLED driver may be shielded from light during the inspection processes.
- Although this OLED display module stores the operation state data by the commands and the indication data, when excessive external noise, etc. enters into the module, the internal status may be changed. Therefore, it is necessary to take appropriate measures to suppress noise generation or to protect from influences of noise on the system design.
- We recommend that you construct its software to make periodical refreshment of the operation statuses (re-setting of the commands and re-transference of the display data), to cope with catastrophic noise.
- Resistors, capacitors, and other passive components will have different appearance and color caused by the different supplier.
- Crystalfontz has the right to upgrade and modify the product function.
- The limitation of FPC bending:

